SECONDARY PHARMACODYNAMICS
In silico off-targets analysis for PrP-lowering divalent siRNA 2439-exNA

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Introduction

Oligonucleotide medicines may have off-target effects mediated by Watson-Crick base pairing with undesired target RNAs (hybridization-mediated off-target effects). Because these are species-specific, such off-targets for humans can be evaluated through in silico analysis[1].

Methods

The reverse complement of bases 2-17 of the 2439-exNA sequence — CACTTTGTGAGTATTC — was searched in NCBI Nucleotide BLAST (https://blast.ncbi.nlm.nih.gov/Blast.cgi) with the following settings:

- Database: Standard databases (nr etc.): > RefSeq Select RNA sequences (refseq_select)
- Organism: Homo sapiens (taxid:9606)*
- Optimize for: Highly similar sequences (megablast)
- Max target sequences: 5,000
- Match/mismatch scores: 1,-1
- All other settings default

*For the nonclinical species searches, we used RefSeq Select for Mus musculus (taxid:10090), Rattus norvegicus (taxid:10116), and RefSeq RNA (no RefSeq Select available) for Canis familiaris (taxid:9615).

The top 100 hits from BLAST were downloaded as an XML file and parsed into tabular form. The table was then filtered to include only matches where the search string was antisense to the target, consistent with the siRNA mechanism.

Results

Human off-targets

The 21-base pair sequence of the antisense strand of 2439-exNA is 5'-UGAAUACUCACAAAGUGCAUU-3'. Base 1 and bases 20-21 were fixed as U and are not designed to be complementary to the target mRNA. It is established [Birmingham 2006] that the 16 bases from 2-17 are most critical for determining activity against the target RNA.This critical region is

GAAUACUCACAAAGUG. The reverse complement of this, CACTTTGTGAGTATTC, is the target sequence present in the coding (sense) strand of *PRNP* DNA corresponding to the 3'UTR of the mRNA.

All hits with <5 mismatches where the search string was antisense to the target RNA are shown in Table 1.

Table 1. Human RefSeq Select BLAST results with <5 mismatches to the reverse complement of 2439-exNA bases 2-17, CACTTTGTGAGTATTC.

gene symbol	RefSeq accession	mismatches	alignment
PRNP	NM_000311	0	CACTTTGTGAGTATTC |||||||||||||||| CACTTTGTGAGTATTC
FURIN	NM_002569	2	CACTTTGTGAGTATT |||||||||| |||| CACTTTGTGATTATT
ZKSCAN3	NM_024493	2	CACTTTGTGAGTAT |||||||||||||| CACTTTGTGAGTAT
ARSG	NM_001267727	3	ACTTTGTGAGTATT ||||||||| |||| ACTTTGTGATTATT
FCN1	NM_002003	3	ACTTTGTGAGTATT |||||||| ||||| ACTTTGTGTGTATT
FLRT2	NM_013231	3	CTTTGTGAGTATTC |||||||| ||||| CTTTGTGAATATTC
GNL1	NM_005275	3	CACTTTGTGAGTAT |||||| ||||||| CACTTTATGAGTAT
GPR63	NM_030784	3	CTTTGTGAGTATTC ||||||||| |||| CTTTGTGAGAATTC
GPRC6A	NM_148963	3	CTTTGTGAGTATTC |||||||| ||||| CTTTGTGATTATTC
LINGO2	NM_001258282	3	CTTTGTGAGTATTC ||||| |||||||| CTTTGGGAGTATTC
NOTCH2NLA	NM_203458	3	ACTTTGTGAGTATT ||||| |||||||| ACTTTTTGAGTATT
NOTCH2NLB	NM_001364008	3	ACTTTGTGAGTATT ||||| |||||||| ACTTTTTGAGTATT
NOTCH2NLC	NM_001364013	3	ACTTTGTGAGTATT ||||| |||||||| ACTTTTTGAGTATT
PNPLA2	NM_020376	3	CACTTTGTGAGTAT ||||||||| |||| CACTTTGTGTGTAT
PRMT3	NM_005788	3	CTTTGTGAGTATTC |||||||| ||||| CTTTGTGACTATTC
SCD	NM_005063	3	CTTTGTGAGTATTC |||||| |||||| CTTTGTGTGTATTC
SLC46A3	NM_181785	3	CACTTTGTGAGTAT ||||||||| |||| CACTTTGTGCGTAT
ABCA1	NM_005502	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGACTATT
ADAMTS5	NM_007038	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
AIFM1	NM_004208	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT
AP5M1	NM_018229	4	CACTTTGTGAGTA |||| |||||||| CACTCTGTGAGTA

gene symbol	RefSeq accession	mismatches	alignment
APOL6	NM_030641	4	TTTGTGAGTATTC ||||| ||||||| TTTGTAAGTATTC
ARID4A	NM_002892	4	CACTTTGTGAGTA ||||| ||||||| CACTTGGTGAGTA
ARMC8	NM_001363941	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
BTRC	NM_033637	4	CTTTGTGAGTATT |||| |||||||| CTTTATGAGTATT
CAMSAP2	NM_203459	4	CACTTTGTGAGTA ||||||| ||||| CACTTTGAGAGTA
CEP57L1	NM_001271852	4	ACTTTGTGAGTA |||||||||||| ACTTTGTGAGTA
CHRNA7	NM_000746	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
CPSF6	NM_007007	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
CRIPT	NM_014171	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
CRY2	NM_021117	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
DIXDC1	NM_001037954	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGATTATTC
DNAJB14	NM_001031723	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
FBXW2	NM_012164	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
FERRY3	NM_020374	4	ACTTTGTGAGTAT ||||||| ||||| ACTTTGTTAGTAT
FGF10	NM_004465	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGATTATTC
FRMPD4	NM_001368397	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGTGTAT
GCNT4	NM_001366737	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
GRIN2B	NM_000834	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
HIVEP1	NM_002114	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
HOXC8	NM_022658	4	CTTTGTGAGTATT ||||| ||||||| CTTTGAGAGTATT
HSPA4L	NM_014278	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
IL2RB	NM_000878	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGATTATT
INTS7	NM_015434	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
IST1	NM_001270975	4	CTTTGTGAGTATT |||| |||||||| CTTTCTGAGTATT
MALSU1	NM_138446	4	CACTTTGTGAGTA |||||||| |||| CACTTTGTTAGTA
MCTP2	NM_001385001	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGCGTAT

gene symbol	RefSeq accession	mismatches	alignment
MGAT5	NM_002410	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
MLXIP	NM_014938	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGTGTAT
MMP16	NM_005941	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGGGTAT
MRAP2	NM_138409	4	CTTTGTGAGTATT ||||| ||||||| CTTTGGGAGTATT
MTMR9	NM_015458	4	TTTGTGAGTATTC ||||| ||||||| TTTGTAAGTATTC
NCAPG	NM_022346	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT
NCK1	NM_001291999	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGAATTC
OR10G7	NM_001004463	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT
PAG1	NM_018440	4	ACTTTGTGAGTAT ||||| ||||||| ACTTTTTGAGTAT
PCTP	NM_021213	4	ACTTTGTGAGTAT ||||| ||||||| ACTTTATGAGTAT
PLEKHA3	NM_019091	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
POLDIP3	NM_032311	4	CTTTGTGAGTATT ||||||| ||||| CTTTGTGTGTATT
PSMD5	NM_005047	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGTGTAT
RAB22A	NM_020673	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
RIOK1	NM_031480	4	CTTTGTGAGTATT ||||||| ||||| CTTTGTGTGTATT
SCFD2	NM_152540	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGGGTAT
SCN8A	NM_001330260	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGATTATTC
SLC7A3	NM_032803	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT
SMCHD1	NM_015295	4	TTTGTGAGTATTC |||| |||||||| TTTGGGAGTATTC
SUGT1	NM_006704	4	ACTTTGTGAGTAT ||||||| ||||| ACTTTGTCAGTAT
TF	NM_001063	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
TMEM255A	NM_001104544	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGACTATT
TMEM59L	NM_012109	4	CTTTGTGAGTATT |||| |||||||| CTTTCTGAGTATT
TP53INP2	NM_021202	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
TRIM24	NM_015905	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGACTATTC
VNN2	NM_004665	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGTGTAT

gene symbol	RefSeq accession	mismatches	alignment
VPS26C	NM_006052	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGATTATTC
ZMIZ1	NM_020338	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT
ZNF567	NM_001322917	4	ACTTTGTGAGTAT |||| |||||||| ACTTAGTGAGTAT
ZNF98	NM_001098626	4	CTTTGTGAGTATT ||||||| ||||| CTTTGTGGGTATT

In order to place any adverse findings from nonclinical studies in context, we also performed the same search for off-targets in mouse, dog, and rat.

Mouse off-targets

Table 2. Mouse RefSeq Select BLAST results with <5 mismatches to the reverse complement of 2439-exNA bases 2-17, CACTTTGTGAGTATTC.

gene symbol	RefSeq accession	mismatches	alignment
Furin	NM_011046	1	CACTTTGTGAGTATTC |||||||||| ||||| CACTTTGTGATTATTC
Ssh2	NM_177710	2	ACTTTGTGAGTATT |||||||||||||| ACTTTGTGAGTATT
Chmp4c	NM_025519	3	CACTTTGTGAGTAT |||||||| ||||| CACTTTGTAAGTAT
Fyco1	NM_001110253	3	ACTTTGTGAGTATT |||||| ||||||| ACTTTGAGAGTATT
Hcfc2	NM_001368720	3	CTTTGTGAGTATTC ||||| |||||||| CTTTGAGAGTATTC
Parp14	NM_001039530	3	ACTTTGTGAGTATT ||||| |||||||| ACTTTTTGAGTATT
Pramel5	NM_001085418	3	CACTTTGTGAGTAT |||||| ||||||| CACTTTCTGAGTAT
Steap2	NM_001103156	3	CTTTGTGAGTATTC ||||||| |||||| CTTTGTGTGTATTC
Tle1	NM_011599	3	ACTTTGTGAGTATT |||||||| ||||| ACTTTGTGTGTATT
Zfp185	NM_009549	3	CACTTTGTGAGTAT ||||||||| |||| CACTTTGTGTGTAT
Zfp871	NM_172458	3	CTTTGTGAGTATT |||||||||||||

			CTTTGTGAGTATT
Adamts5	NM_011782	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
Arid4a	NM_001081195	4	CACTTTGTGAGTA ||||| ||||||| CACTTGGTGAGTA
Atp1a3	NM_001290469	4	CTTTGTGAGTAT |||||||||||| CTTTGTGAGTAT
Ccdc15	NM_001081429	4	ACTTTGTGAGTA |||||||||||| ACTTTGTGAGTA
Ccdc186	NM_170757	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT
Cdon	NM_021339	4	TTTGTGAGTATTC ||||| ||||||| TTTGTCAGTATTC
Cept1	NM_133869	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGATTATTC
Col25a1	NM_029838	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGTGTAT
Cyld	NM_173369	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
Dach2	NM_033605	4	CTTTGTGAGTATT ||||| ||||||| CTTTGGGAGTATT
Eppin	NM_029325	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGCATTC
Grin2b	NM_001363750	4	CTTTGTGAGTATT ||||| ||||||| CTTTGAGAGTATT
Hdgfl3	NM_013886	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
Hoxc8	NM_010466	4	CTTTGTGAGTATT ||||| ||||||| CTTTGAGAGTATT
Khdc1a	NM_183322	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGAATTC
Khdc1c	NM_001033904	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGAATTC
Lrriq1	NM_001163559	4	CACTTTGTGAGTA |||||||| |||| CACTTTGTCAGTA
Lyst	NM_010748	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
Map2k3	NM_008928	4	CTTTGTGAGTATT ||||||| ||||| CTTTGTGGGTATT
Mmut	NM_008650	4	TTTGTGAGTATTC |||| |||||||| TTTGGGAGTATTC
Nat8f2	NM_053096	4	ACTTTGTGAGTAT ||||||| ||||| ACTTTGTCAGTAT
Ncapg	NM_019438	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT
Ndufa9	NM_025358	4	TTTGTGAGTATTC ||||||| |||||

			TTTGTGAATATTC
Nfib	NM_001113209	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGATTATTC
Nr2c1	NM_011629	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
Olfml1	NM_172907	4	CACTTTGTGAGTA ||||||| ||||| CACTTTGGGAGTA
Olfr207	NM_001011792	4	CTTTGTGAGTATT |||| |||||||| CTTTATGAGTATT
Olfr209	NM_207551	4	CTTTGTGAGTATT |||| |||||||| CTTTATGAGTATT
Otud5	NM_138604	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
Pctp	NM_008796	4	ACTTTGTGAGTAT ||||| ||||||| ACTTTATGAGTAT
Ppcs	NM_026494	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGATTATT
Pramel11	NM_001406740	4	ACTTTGTGAGTA |||||||||||| ACTTTGTGAGTA
Pramel21	NM_001126324	4	ACTTTGTGAGTA |||||||||||| ACTTTGTGAGTA
Pramel24	NM_001085412	4	ACTTTGTGAGTA |||||||||||| ACTTTGTGAGTA
Ptpn2	NM_001127177	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
Rasl12	NM_001033158	4	CACTTTGTGAGTA |||||||| |||| CACTTTGTTAGTA
Ric1	NM_001081319	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
Shtn1	NM_001114312	4	CACTTTGTGAGTA |||||||| |||| CACTTTGTAAGTA
Strip2	NM_177204	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT
Tbl1xr1	NM_030732	4	ACTTTGTGAGTAT ||||| ||||||| ACTTTTTGAGTAT
Tfam	NM_009360	4	TTTGTGAGTATTC |||| |||||||| TTTGAGAGTATTC
Timm21	NM_025969	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
Tmem255a	NM_172930	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGACTATT
Trmt2b	NM_172540	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
Ube2k	NM_016786	4	CACTTTGTGAGTA |||||||| |||| CACTTTGTAAGTA
Ube3a	NM_011668	4	CTTTGTGAGTATT |||| ||||||||

			CTTTTTGAGTATT
Vezt	NM_172538	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
Vmn2r76	NM_001384930	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGACTATT
Vps29	NM_019780	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGTGTAT
Yme1l1	NM_013771	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGTGTAT
Zfp174	NM_001081217	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGATTATT
Zfp617	NM_133358	4	CTTTGTGAGTAT |||||||||||| CTTTGTGAGTAT
Zfp948	NM_001002008	4	CTTTGTGAGTATT ||||||| ||||| CTTTGTGTGTATT
Zkscan2	NM_001081329	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGCATTC
Zmiz1	NM_001310666	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT

Rat off-targets

Table 3. Rat RefSeq Select BLAST results with <5 mismatches to the reverse complement of 2439-exNA bases 2-17, CACTTTGTGAGTATTC.

gene symbol	RefSeq accession	mismatches	alignment
Furin	XM_008759555	1	CACTTTGTGAGTATTC |||||||||| ||||| CACTTTGTGATTATTC
Msr1	NM_001191939	2	ACTTTGTGAGTATTC |||||| |||||||| ACTTTGGGAGTATTC
Rad51	NM_001109204	2	CACTTTGTGAGTATT ||||||| ||||||| CACTTTGGGAGTATT
B3galt2	NM_001109492	3	TTTGTGAGTATTC ||||||||||||| TTTGTGAGTATTC
Dgke	NM_001394358	3	ACTTTGTGAGTATT ||||| |||||||| ACTTTTTGAGTATT
Dtx3l	NM_001109053	3	CACTTTGTGAGTAT |||||| ||||||| CACTTTTTGAGTAT
Fam120a	NM_001191816	3	ACTTTGTGAGTATT ||||||||| |||| ACTTTGTGATTATT
Hcfc2	NM_001008357	3	CTTTGTGAGTATTC ||||| |||||||| CTTTGAGAGTATTC

gene symbol	RefSeq accession	mismatches	alignment
LOC120097096	XM_063274845	3	CTTTGTGAGTATT ||||||||||||| CTTTGTGAGTATT
LOC134485275	XM_063281052	3	ACTTTGTGAGTATT ||||||||| |||| ACTTTGTGATTATT
Lingo2	XM_039109831	3	CTTTGTGAGTATTC ||||| |||||||| CTTTGGGAGTATTC
Ncapg	NM_001398911	3	CTTTGTGAGTATT ||||||||||||| CTTTGTGAGTATT
Phactr2	XM_017589098	3	CTTTGTGAGTATTC |||||| ||||||| CTTTGTTAGTATTC
Stk35	NM_001107773	3	CACTTTGTGAGTAT ||||||||| |||| CACTTTGTGTGTAT
Atp13a4	NM_001191658	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGGGTAT
Ccdc14	NM_001427244	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
Cept1	XM_006233116	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGATTATTC
Dixdc1	NM_001395147	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGATTATTC
Dppa3l2	XM_039096319	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT
Fahd1	NM_001024991	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGTGTAT
Gpr174	XM_006257164	4	CTTTGTGAGTATT ||||||| ||||| CTTTGTGGGTATT
Kansl1l	NM_001431332	4	CACTTTGTGAGTA |||| |||||||| CACTGTGTGAGTA
LOC102546572	NM_001432177	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT
LOC102550654	XM_063274812	4	TTTGTGAGTATTC |||| |||||||| TTTGGGAGTATTC
LOC120101034	XM_039103930	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
LOC120101038	XM_039103931	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
LOC134478816	XM_039096295	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
LOC134479581	XM_063264370	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGTGTAT
LOC134479904	XM_063264625	4	CACTTTGTGAGTA ||||| ||||||| CACTTAGTGAGTA
LOC134482899	XM_063277639	4	CTTTGTGAGTAT |||||||||||| CTTTGTGAGTAT

gene symbol	RefSeq accession	mismatches	alignment
LOC134485799	XM_063282819	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
LOC134486557	XM_063286815	4	ACTTTGTGAGTA |||||||||||| ACTTTGTGAGTA
Lrriq1	XM_063264416	4	CACTTTGTGAGTA |||||||| |||| CACTTTGTCAGTA
Lyst	NM_053518	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
Mfsd8	NM_001393796	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
Mmp24	NM_031757	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGACTATTC
Mmut	NM_001399409	4	TTTGTGAGTATTC |||| |||||||| TTTGGGAGTATTC
Ndufa9	NM_001100752	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
Nr2c1	NM_145780	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
Olr1057	NM_001000072	4	ACTTTGTGAGTAT ||||| ||||||| ACTTTCTGAGTAT
Or5p80	NM_001409148	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGACTATTC
Or6c70	NM_001000704	4	ACTTTGTGAGTAT ||||| ||||||| ACTTTCTGAGTAT
Pappa2	NM_001427778	4	CTTTGTGAGTATT |||| |||||||| CTTTATGAGTATT
Parg	NM_031339	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGAATTC
Pkhd1	XM_008766937	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGACTATTC
Poglut3	NM_001025123	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
Polrmt	NM_001106766	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGCATTC
Ppcs	NM_001039010	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGATTATT
Ric1	NM_001427735	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGAATATTC
Rpl15	NM_139114	4	TTTGTGAGTATTC ||||||| ||||| TTTGTGACTATTC
Shtn1	NM_001079705	4	CACTTTGTGAGTA |||||||| |||| CACTTTGTAAGTA
Smco3l1	XM_039108871	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT

gene symbol	RefSeq accession	mismatches	alignment
Stum	XM_006250416	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
Tarbp1	XM_039098287	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
Tdpoz1-ps14	XM_063282818	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
Tdpoz2l10	XM_039103933	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
Tdpoz2l12	XM_039103934	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
Tdpoz2l3	XM_039103929	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
Tdpoz2l4	XM_003749331	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
Tdpoz2l8	XM_039103936	4	TTTGTGAGTATTC |||||||| |||| TTTGTGAGGATTC
Tead4	NM_001427269	4	CACTTTGTGAGTA |||| |||||||| CACTCTGTGAGTA
Tead4l1	XM_006237423	4	CACTTTGTGAGTA |||| |||||||| CACTCTGTGAGTA
Tmed2	NM_031722	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGATTATT
Tmem255a	NM_182822	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGACTATT
Tubg1	NM_145778	4	CTTTGTGAGTATT ||||| ||||||| CTTTGAGAGTATT
Ubap1l	XM_039082471	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
Ube2k	NM_001106006	4	CACTTTGTGAGTA |||||||| |||| CACTTTGTAAGTA
Ube3a	NM_001191837	4	CTTTGTGAGTATT |||| |||||||| CTTTTTGAGTATT
Vash1	NM_001427235	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
Vom1r101	NM_173112	4	ACTTTGTGAGTAT |||||||| |||| ACTTTGTGTGTAT
Vom2r42	NM_001099511	4	CTTTGTGAGTAT |||||||||||| CTTTGTGAGTAT
Zmiz1	NM_001108393	4	CTTTGTGAGTATT |||||||| |||| CTTTGTGAATATT

Dog off-targets

Table 4. Dog RefSeq BLAST results with <5 mismatches to the reverse complement of 2439-exNA bases 2-17, CACTTTGTGAGTATTC.

gene symbol	RefSeq accession	mismatches	alignment
AKAP11	XM_038569639	1	CACTTTGTGAGTATTC |||||||||| ||||| CACTTTGTGATTATTC
CACNB4	XM_038459057	2	ACTTTGTGAGTATTC ||||||| ||||||| ACTTTGTAAGTATTC
FURIN	XM_038532884	2	CACTTTGTGAGTATT |||||||||| |||| CACTTTGTGATTATT
LOC111097308	XR_005354431	2	CTTTGTGAGTATTC |||||||||||||| CTTTGTGAGTATTC
LOC111098102	XR_005434232	2	CACTTTGTGAGTATT ||||| |||||||| CACTTAGTGAGTATT
LOC119864503	XM_038565207	2	ACTTTGTGAGTATTC ||||||| ||||||| ACTTTGTAAGTATTC
MAP2K6	XM_038546746	2	CACTTTGTGAGTATT ||||| |||||||| CACTTCGTGAGTATT
MMP16	XM_038579700	2	CACTTTGTGAGTAT |||||||||||||| CACTTTGTGAGTAT
TMEM170B	XM_038584175	2	CACTTTGTGAGTATT |||| ||||||||| CACTGTGTGAGTATT
VPS35	XM_038557981	2	CACTTTGTGAGTATT |||||| |||||||| CACTTTTTGAGTATT
WDR12	XM_038447672	2	ACTTTGTGAGTATTC |||||||| |||||| ACTTTGTGGGTATTC
WNK3	XM_038587521	2	CACTTTGTGAGTATT |||||||||| |||| CACTTTGTGAATATT
LOC111097008	XR_005363200	3	CTTTGTGAGTATT ||||||||||||| CTTTGTGAGTATT
LOC119870875	XM_038527581	3	CTTTGTGAGTATT ||||||||||||| CTTTGTGAGTATT
LOC119871444	XR_005357797	3	CTTTGTGAGTATT ||||||||||||| CTTTGTGAGTATT
LOC119875986	XM_038527580	3	CTTTGTGAGTATT ||||||||||||| CTTTGTGAGTATT
LOC119879597	XM_038655595	3	CTTTGTGAGTATT ||||||||||||| CTTTGTGAGTATT
LOC609690	XR_005364533	3	ACTTTGTGAGTAT ||||||||||||| ACTTTGTGAGTAT

ADAMTS5	XM_038581284	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
ARMC8	XM_038571002	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
FBXO16	XM_038435317	4	CTTTGTGAGTAT |||||||||||| CTTTGTGAGTAT
GRAMD2A	XM_038580868	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
HIVEP1	XM_038584179	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
LEPR	XM_038664087	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
LOC102151722	XR_005382713	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
LOC102152102	XM_038655554	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
LOC102153134	XR_005435474	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
LOC102155123	XR_293955	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
LOC102155985	XR_005383223	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
LOC106560064	XR_005374327	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
LOC111093240	XR_002621122	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
LOC111094776	XR_005319037	4	CTTTGTGAGTAT |||||||||||| CTTTGTGAGTAT
LOC111098802	XR_005369562	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
LOC119867345	XR_005383404	4	ACTTTGTGAGTA |||||||||||| ACTTTGTGAGTA
LOC119868529	XR_005386490	4	CTTTGTGAGTAT |||||||||||| CTTTGTGAGTAT
LOC119870783	XR_005353846	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT
LOC119876904	XR_005368225	4	ACTTTGTGAGTA |||||||||||| ACTTTGTGAGTA
MPPED2	XR_005373414	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
NEDD4L	XM_038525922	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
RABGEF1	XM_038539000	4	ACTTTGTGAGTA |||||||||||| ACTTTGTGAGTA
RFLNB	XM_038548444	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT

RNF125	XM_038672394	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
RP2	XM_038449860	4	ACTTTGTGAGTA |||||||||||| ACTTTGTGAGTA
TPGS2	XM_038543469	4	TTGTGAGTATTC |||||||||||| TTGTGAGTATTC
VPS53	XR_005364750	4	CACTTTGTGAGT |||||||||||| CACTTTGTGAGT
ZDHHC20	XM_038573488	4	TTTGTGAGTATT |||||||||||| TTTGTGAGTATT

Conclusions

In vivo in the CNS, siRNAs possess reduced activity against sites with 1 mismatch, and virtually no activity against sites with ≥2 mismatches[2,3]. Some experiments, particularly in cultured cells, have shown off-target activity enriched for transcripts with copies of sequence complementary to only the seed region (bases 2-8)[4], but in the CNS, the seed complement does not appear to be enriched for off-target effects[5]. Overall, evidence indicates that transcripts with ≥2 mismatches are unlikely to experience any off-target lowering. In the human transcriptome, all non-*PRNP* genes have at least 2 mismatches to 2439-exNA, thus minimizing the risk of unwanted off-target effects. The nonclinical species utilized in this IND — mouse, rat, and dog — all possess at least 1 gene with only a single mismatch to 2439-exNA.

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